Document Identifier: **PMC8728528**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Consent: Informed consent was obtained from all the patients, and sample collection complied with the approved guidelines of Ethics Committee of the Kunming Medical University. GBC TNM (Tumor-Node-Metastasis) stages were evaluated according to the 8th edition of the American Joint Committee on Cancer staging system (AJCC/UICC TNM, 8th edition).
IRB: Informed consent was obtained from all the patients, and sample collection complied with the approved guidelines of Ethics Committee of the Kunming Medical University. GBC TNM (Tumor-Node-Metastasis) stages were evaluated according to the 8th edition of the American Joint Committee on Cancer staging system (AJCC/UICC TNM, 8th edition).
Field Sample Permit: Cell culture GBC-SD cells were purchased from the Cell Bank of Type Culture Collection of Chinese Academy of Sciences and cultured in Dulbecco's Modified Eagle Medium (DMEM)/ F12 (Gibco, Grand Island, NY, USA) supplemented with 10% fetal bovine serum (FBS, BI, Biological Industries, Beit Haemek, Israel), and 100 U/mL penicillin and 100 µg/mL streptomycin (BI, Beit Haemek,
IACUC: Animal care and all procedures were approved and under the supervision of Yunnan Provincial Laboratory Animal Welfare Committee.
Inclusion and Exclusion Criteria
not detected.
Attrition
Paraffin-embedded tissues from 108 cases of chronic cholecystitis, 92 cases of dysplasia, and 72 cases of GBC were obtained and cut into 4-mm sections for further analysis.
Sex as a biological variable
Tumor xenografts in nude mice Four-week-old male BALB/c nude mice were purchased from the Hu Nan Animal Center.
Subject Demographics
Age: Tumor xenografts in nude mice Four-week-old male BALB/c nude mice were purchased from the Hu Nan Animal Center.
Randomization

After 15 days of tumor growth, the mice were randomly assigned into the treatment or NS group.
Blinding
The results were evaluated and recorded by two pathologists blinded to the case information.
Power Analysis
not detected.
Replication
Statistical analysis Data were acquired from experiments conducted in triplicate.
Number: Statistical analysis Data were acquired from experiments conducted in triplicate.
Cell Line Authentication
Authentication: Cell lines were authenticated by Short Tandem Repeat profiling before use in the experiments .

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Primary antibodies were used for immunoblotting: anti-OLFM4 antibody (Cat No. 14369D , Cell Signaling Technology , Danvers , Massachusetts , USA) , caspase-3/ p17/p19 (Cat No. 66,470–2-Ig , Proteintech , Rosemont , IL , USA)	anti-OLFM4		
	caspase-3/p17/ p19		
Experimental Models: Organisms/Strains			
Tumor xenografts in nude mice Four-week-old male BALB/c nude mice were purchased from the Hu Nan Animal Center.	BALB/c nude		
Software and Algorithms			
Bioinformatic analysis of differential expression levels of mRNA and lncRNA from microarrays Gene set enrichment analysis (GSEA) was performed to identify potential biological mechanisms associated with OLFM4 expression in GBC sand normal paracarcinoma tissues.	Gene set enrichment analysis		Suggestion: Gene Set Enrichment Analysis (RRID:SCR_003199)(link)
GSEA software and gene sets were downloaded from https://www.gsea-msigdb.org/.	GSEA		
The cells were trypsinized and washed twice with PBS , then the cells were resuspended in 100 μL of serum free DMEM/F12 medium (Gibco) and seeded into the upper chamber .	Gibco		
Then the cells were quantified using ImageJ software (National Institutes of Health , Bethesda , MD , USA) .	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
Cells were stained with propidium iodide solution for 20 min in a 37 °C incubator and then analyzed on a flow cytometer (Beckman Coulter , Brea , CA , USA) , and data were analyzed and generated using Modfit software (http://www.vsh.com/products/mflt/ index.asp).	http:// www.vsh.com/ products/mflt/ index.asp		
Statistical analyses were performed using SPSS version 23.0 (IBM Corp., Armonk, NY, USA) or Graph Pad Prism version 8.0 (GraphPad Inc., San Diego, CA USA).	SPSS		Suggestion: SPSS, RRID:SCR_002865(link)
	Graph Pad Prism		
	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Differentially expressed genes (DEGs) were screened out under the criterion of absolute fold change (FC) ≥ 2 and $p < 0.05$ (Student's t-test).	t-test
Clinicopathological associations with OLFM4 H-scores were analyzed using chi-square tests.	chi-square tests
Survival curves were generated using log-rank tests.	log-rank tests
Two-groups comparisons were made with Student's t-tests, and comparison for variables in more than two groups was using one- or two-way analysis of variance (ANOVA).	t-tests
	analysis of variance
	ANOVA
Oligonucleotides	
The interference sequence targeting human OLFM4 complementary DNA is shOLFM4–1: CCGGAGTGCAGAGCATTAACTATA ACTCGAGTTATAGTTAATGCTCTG CACTTTTTTG; shOLFM4–2: CCGGCCCTAATGCTGCCTATAATA ACTCGAGTTATTATAGGCAGCATT AGGGTTTTTG; and a scramble shRNA was included as negative control (shNC): 5#-TTCTCCGAACGTGTCACGT-3 # .	CCGGAGTGCAGAGCATTAACTATA ACTCGAGTTATAGTTAATGCTCTG CACTTTTTTG
	CCGGCCCTAATGCTGCCTATAATA ACTCGAGTTATTATAGGCAGCATT AGGGTTTTTG
	5#-TTCTCCGAACGTGTCACGT-3 #

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq